Basic Principles of Medicine 1
Module : Foundations to Medicine
FTM Lecture No: 49

Lecture Title

DNA Repair:

DNA Damage and Mutation

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Objectives : DNA repair

SOM.MKIII.BPM1.1.FTM.3.GNET.0147	Analyze ways that DNA can be damaged, the cellular mechanisms that are employed in an attempt to fix the damage, and the long- and short-term consequences of mutation accumulation
SOM.MKIII.BPM1.1.FTM.3.GNET.0148	Analyze genetic disorders associated with defects of DNA repair including Bloom syndrome, Lynch syndrome (HNPCC), xeroderma pigmentosa, ataxia telangiectasia, BRACA 1/2 hereditary breast and ovarian cancer
SOM.MKIII.BPM1.1.FTM.3.GNET.0149	Discuss why some areas of the genome are more prone to mutation than other areas; such as repetitive regions, low copy repeats, and 5-methyl cytosine

Recommended Reading

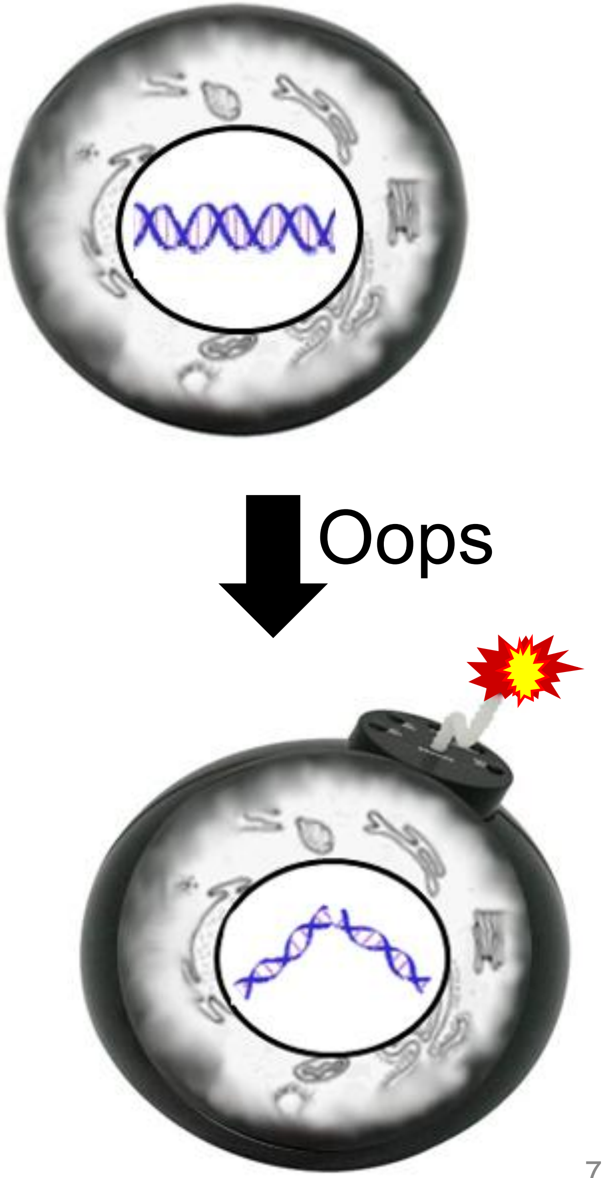
- Korf & Irons: Clinical Snapshot: DNA Repair Disorders; p30
- Korf and Irons text 4th edition-Information in Chapter 2 associated with DNA repair; see in particular, figures 2.1, 2.2, 2.4, 2.5, 2.6, 2.10, 2.11, 2.13, note that table 2.1 includes many disorders that are not discussed in this course, pay attention to the disorders that we discuss.
- Lippincott's Illustrated Reviews: Biochemistry, 8th Edition Chapter 30

DNA Damage

- DNA is damaged thousands of times per cell per day
- Due to many factors:

  -including basic chemistry of the nucleic acids

  -external things like UV light, chemicals, radiation, pH, smoking, *etc. etc.*



Allowing such cells to replicate is "bad"

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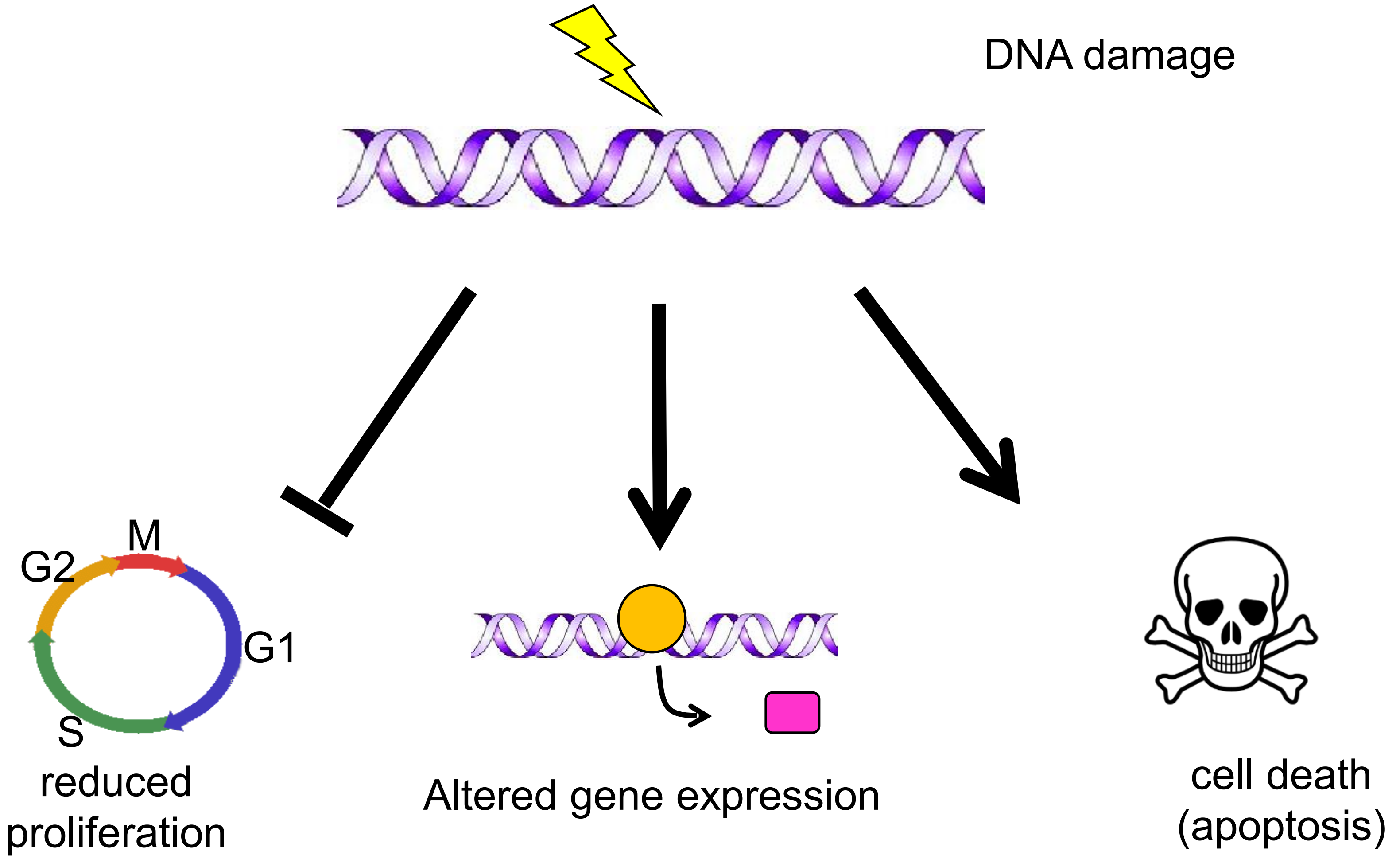
DNA Damage,

Short Term Consequences



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DNA Damage, Long Term Consequences

aging

diseases especially cancer

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Repair of DNA Damage

DNA damage is common

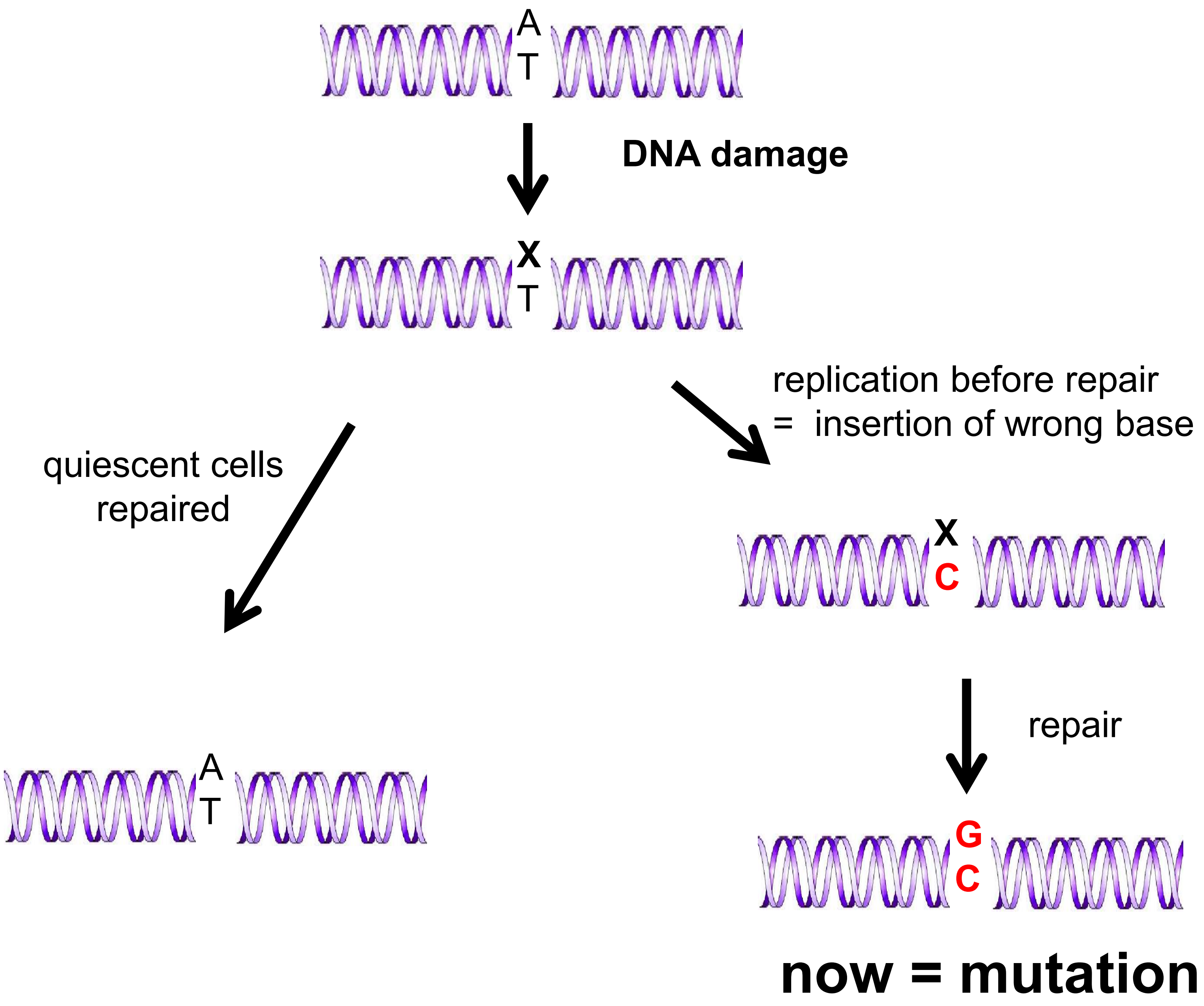
Normal metabolism causes ~10^4 DNA adducts/cell/day through generation of endogenous oxidants (*e.g*. hydrogen peroxide, superoxide, *etc.).*

it can be recognized and efficiently repaired

But

Trying to replicate through it can convert DNA damage into mutations

Important to distinguish between DNA damage and DNA mutation



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Molecular basis of mutation

• mutations can be spontaneous or induced

• induced mutations are in addition to the rate of spontaneous mutation

Two classes of Spontaneous mutation

Errors of replication

= mistakes made during replication

-only occurs during S phase of cell division

Spontaneous lesions

= chemical changes that occur spontaneously

-occurs in resting cell

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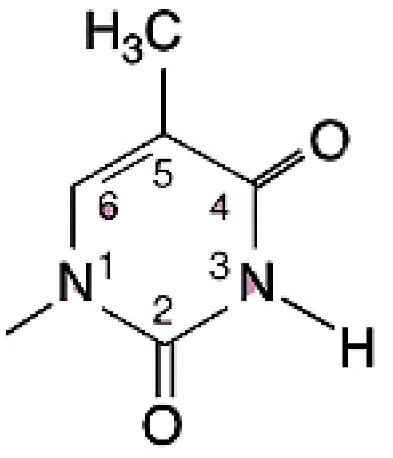
Errors of Replication

Wrong base is incorporated by DNA polymerase

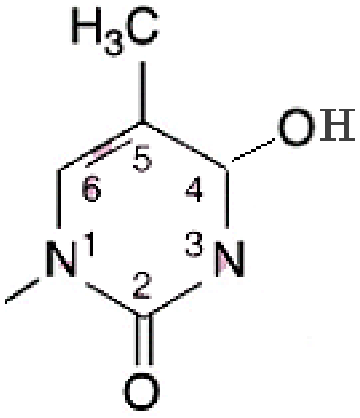
-due to chemistry of the nucleotides

Tautomerism = the ability of certain chemicals to exist as a mixture of two interconvertible isomers



Thymine

pairs with A



Thymine

rare enol form now pairs with G

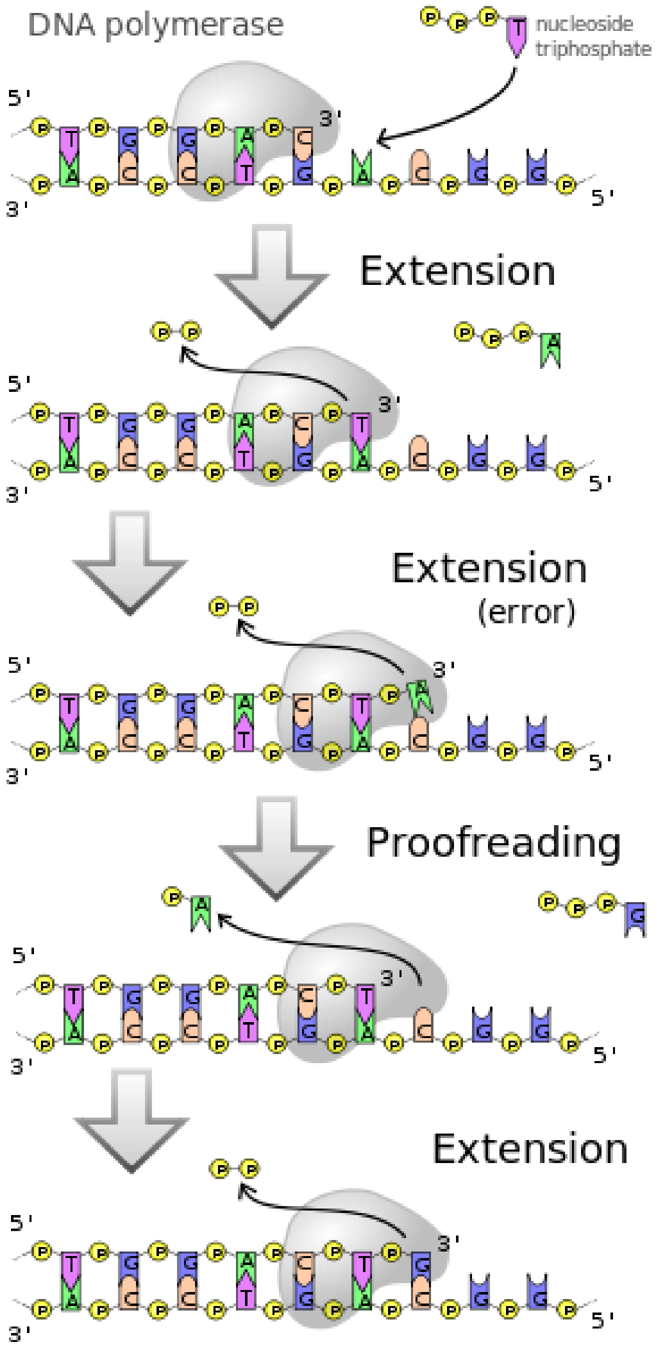
Proof-reading

Replication is very accurate but errors occur

DNA polymerase error rate =1/100,000 bases

Actual error rate = 1 in 10,000,000

DNA Pol enzyme has 5’-3’ polymerase activity AND 3’ to **5’ proof reading exonuclease activity**



Stop and think about it: In which stage of the cell cycle is proof-reading occurring?

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Bloom syndrome

Autosomal recessive disorder

Defect in a gene encoding a DNA helicase enzyme

-required for replication repair, recombination

Characteristics:

-smaller than average
-narrow chin, prominent nose and ears
-facial rash (pigment and dilated blood vessels) upon exposure to sun (sometimes called a 'butterfly' rash
-often get diabetes and have neurological, lung and immune system deficiencies

-**Chromosomal instability resulting in many chromosomal breaks and sister chromatid exchanges**
-Higher risk of a broad range of cancer types

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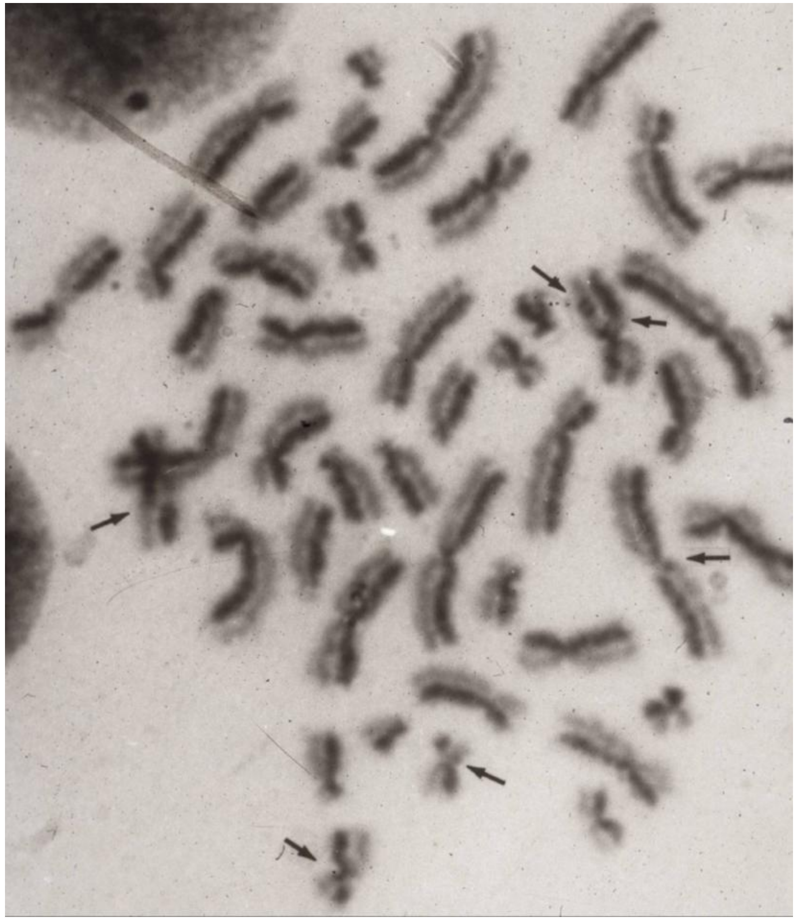
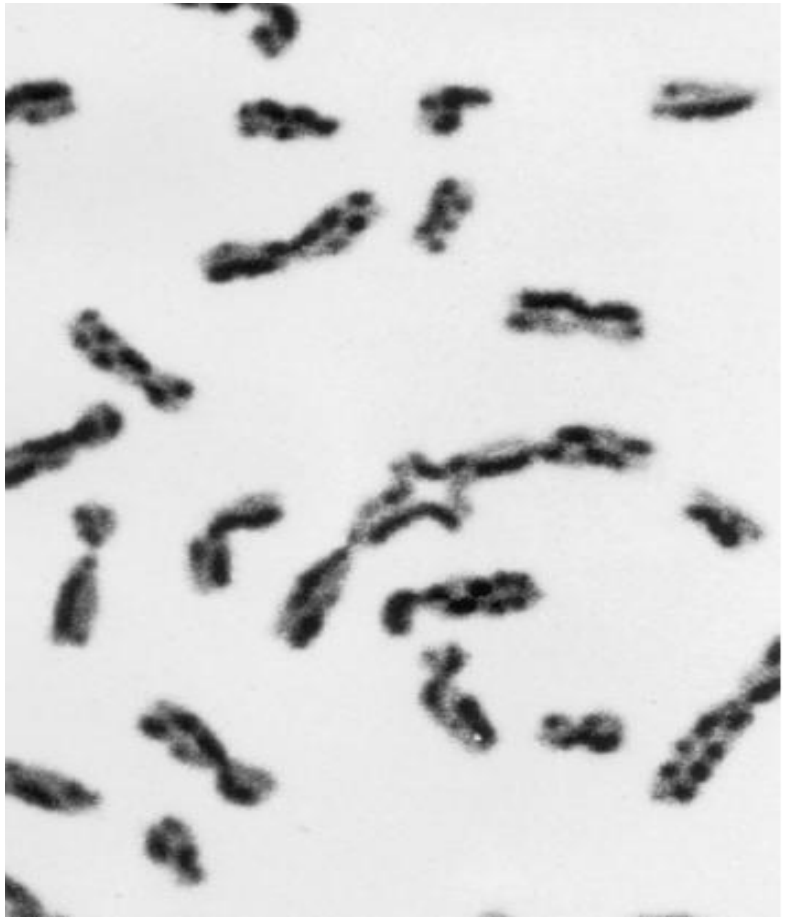
Increased sister chromatid exchange in Bloom syndrome





Stop and think about it: why would a mutation in a helicase enzyme lead to increased chromosome breakage?

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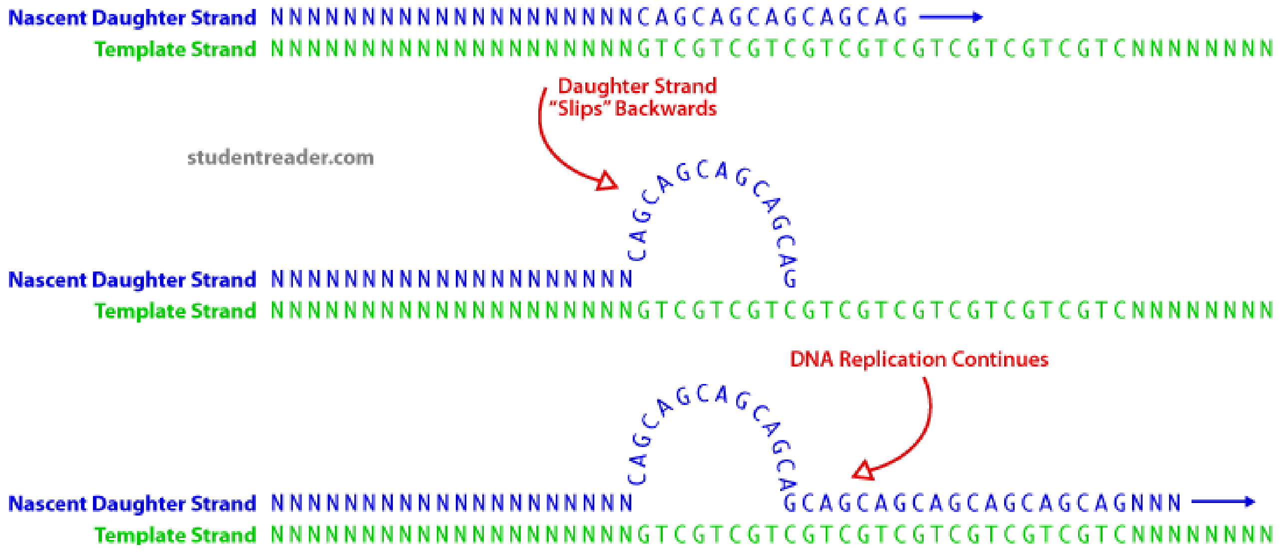
Frameshift mutations

-tend to occur at positions where there are base repeats (*e.g.* GTCGAAAAACTCA)

-thought to result from 'slipping' of DNA polymerases during replication of these repeats

-DNA loops or kinks at these points and one or more bases are not copied or are copied twice

1) DNA polymerase replicates a repeated DNA sequence.
2) DNA polymerase is arrested and dissociates from the newly synthesize strand.
3) The newly synthesized strand separates from the template and realigns with another repeated sequence.
4) DNA polymerase reloads and replication resumes.



Slippage during DNA replication

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Spontaneous lesions

- Changes that occur in a resting cell due to the chemical nature of the DNA

- Extremely common
    - = tens of thousands of mutation events *per* cell *per* day

- Exposure to mutagens such as sunlight, radioactivity, ionizing radiation, specific chemicals may increase the rate of this type of DNA damage

- Three main types of spontaneous DNA damage
    - -depurination
    - -deamination
    - -oxidative damage

1) Depurination

-most common form of spontaneous lesion

~10,000 each day/cell

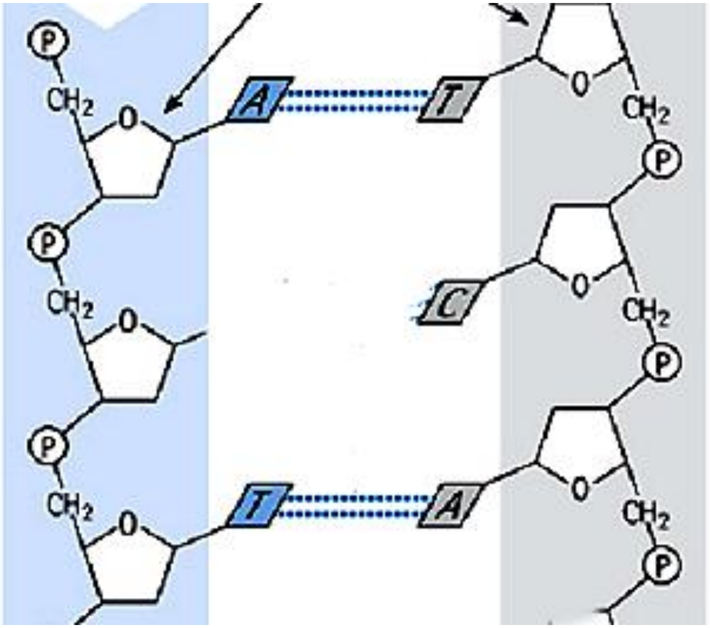
(this is about once every 10 seconds… in every cell of your body)

-breaking of glycosidic bond between base and sugar in purine nucleotides

-sugar-phosphate backbone remains but base is lost

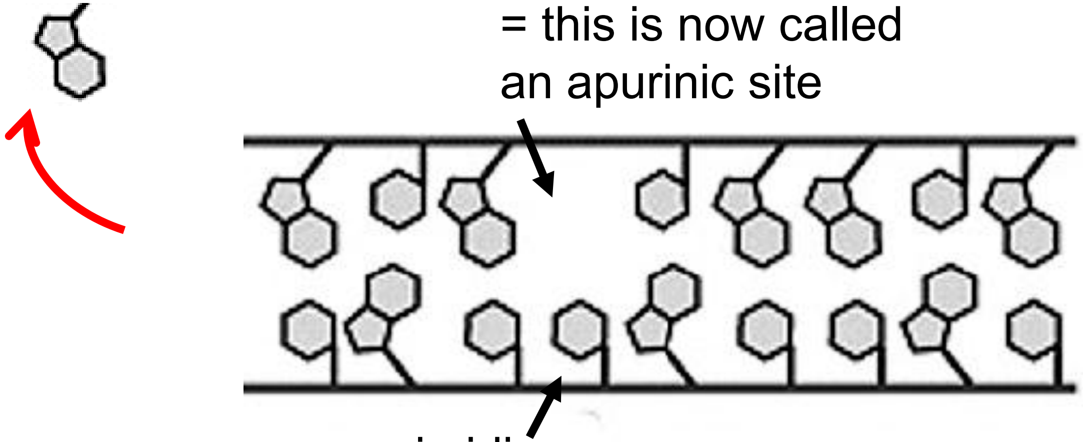
-if it persists through replication then mutation can occur

-these are repaired by a base excision repair mechanism using an enzyme called *AP endonuclease* that nicks the phosphodiester bond at the apurinic site.



Purine base is lost

= this is now called an apurinic site



pyrimidine

2) Deamination

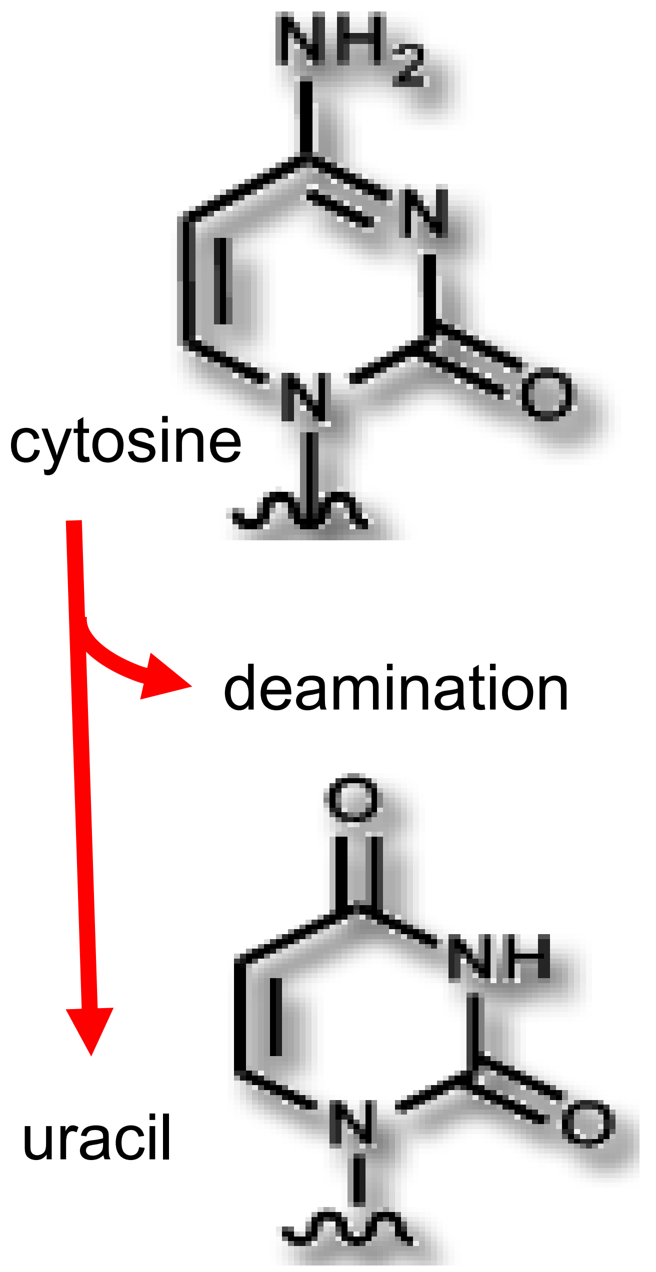
-also very common, ~500/cell/day

-loss of amine group from a base (we can use cytosine as an example)

-cytosine (which base pairs with G) deaminates to form uracil (Uracil would like to pair with A)

-Uracil does not belong in DNA

-Uracil is recognized in DNA and removed by the enzyme *uracil-DNA glycosylase* in a base excision repair mechanism. Uracil-DNA glycosylase breaks the β-N-glycosidic bond of uracil.



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However,

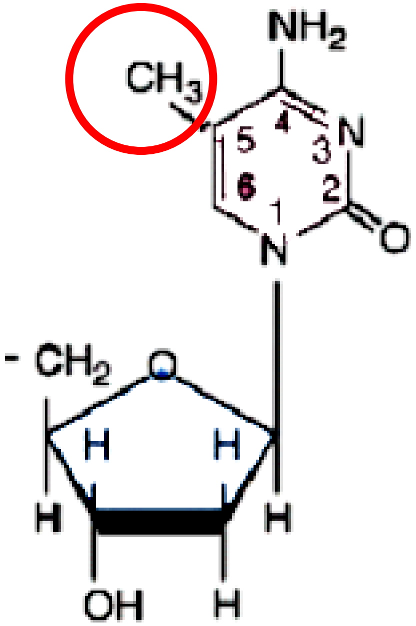
5-methyl cytosine deaminates to form thymine

so this creates a T-G base pair

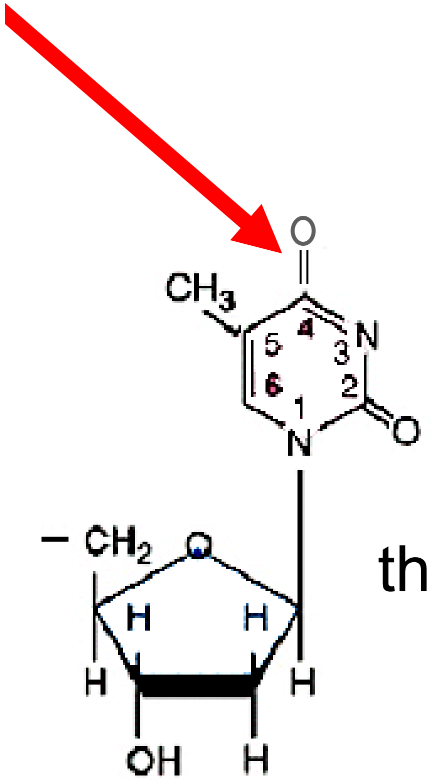
- Both of these bases are normally found in DNA; could be repaired back to a “C” (correct repair) or create the mutation “T”

- So, repair machinery sometimes must “*guess*”

= mutational hotspot



5-methyl cytosine

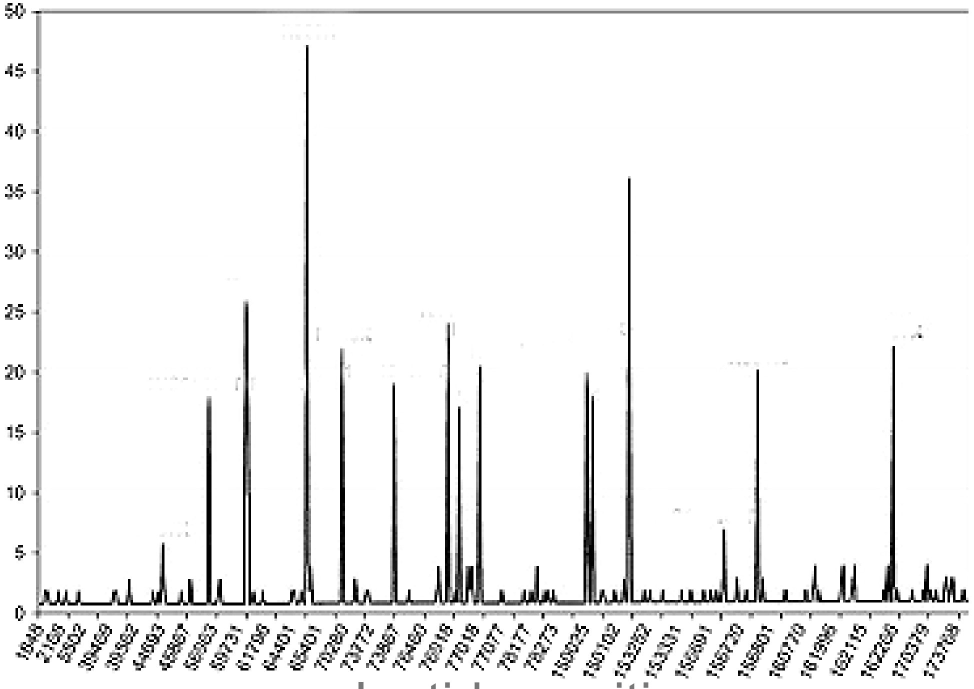


thymine

 Stop and think about it: explain why CG islands are hotspots for mutation

Methylated cytosines are found in mutational hotspots

of mutations



nucleotide position

Found in CpG islands associated with gene promoters

Not all positions are equally mutable, hotspots include things like 5-methyl cytosine or repeated bases (AAAAA)

The point is that you can see the same mutation occurring at a relatively high frequency in specific places in the genome

3) Oxidative damage

-a result of the production of reactive oxidative compounds due to oxidative metabolism

-superoxides, peroxides, *etc.*

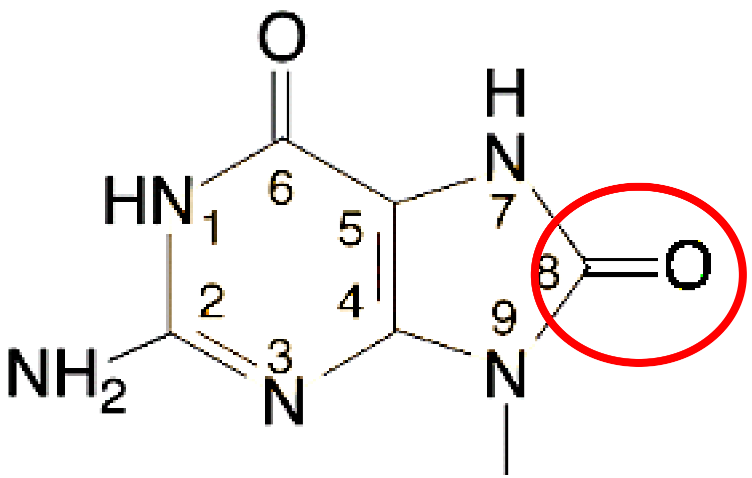
(don't breathe, it is bad for you)

-causes oxidative damage to many parts of cell including addition of oxygen groups to nucleotide bases

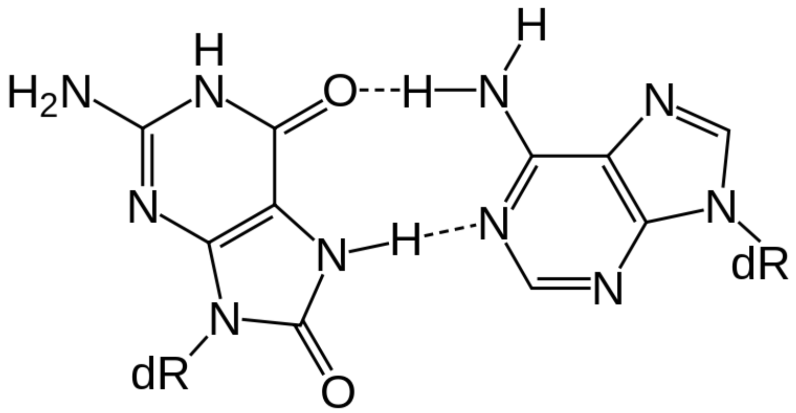
An example would be: 8-oxo-7-hydroxyguanosine (no need to memorize this)

-this oxidized G will mis-pair with **A** and cause a potential **transversion mutation**

-this type of damage is also repaired by a specific glycosylase using the base excision repair mechanism



8-oxo dG



8-oxo G :: A

Mutagens - are mutagenic because they increase the frequency of ‘normal’ mutations (mismatches, depurination, etc.)

There are many, (very many) mutagens in our environment

- Serious
  - UV (from sun or tanning beds)
  - ionizing radiation (from space or the ground)
  - aflatoxin (from moldy grain)
  - benzene
  - formaldehyde
  - mustard gases
  - anything that reacts with DNA
- Not so serious (but should be considered)
  - alcohol (acetaldehyde breakdown product)
  - barbequed hamburgers (benzo[a]pyrene)
  -
  - → anything fun

Ionizing Radiation

-include cosmic rays, γ-rays, x-rays, and radioactive particles (α, β, γ)

-relevant to clinician and research scientists

-but also relevant to everyone as we encounter a surprisingly large amount of radiation
(*e.g. dental x-rays, exit signs, cosmic rays, terrestrial radiation, bananas…*

-high energy particles or rays can cause many types of cellular damage up to and including death

-also causes extensive damage to DNA (base damaging type) **including heritable mutations**

UV light

-generates several deleterious photoproducts

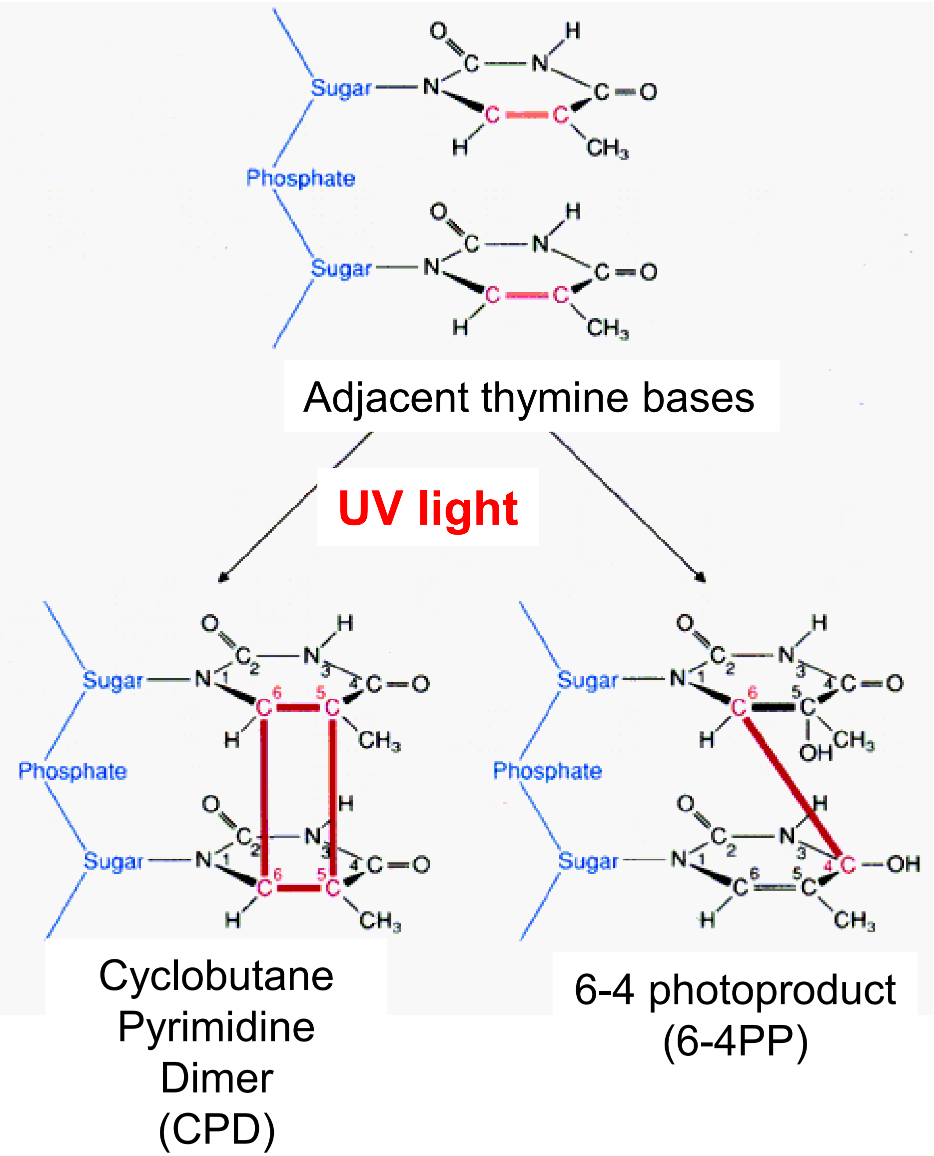
-such as cyclobutane pyrimidine dimers or 6-4 photoproducts

-covalent linkages between adjacent pyrimidine bases on the same strand

-these dimers will interfere with normal base pairing and block replication

These are known as pyrimidine dimers, or thymine dimers

Thymine dimers are recognized by a specific endonuclease that detects distortion in the DNA helix and an excision repair mechanism



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Need for Repair

We are composed of many trillions of cells…

So …. Estimate the number of DNA damage events in a typical person each day

Frequency of errors

• depurination	up to 10,000	/cell/day
• deamination	~500	/cell/day
• methylation errors	>500	/cell/day
• oxidative damage	-up to 500	/cell/day

• UV damage - *depends on exposure*, more UV light, more damage

-UV damage may be extensive as seen by Xeroderma pigmentosum

• *etc., etc.*

• and that doesn't include problems due to all of the man-made mutagens we run into every day

DNA repair defects

NAME	PHENOTYPE	ENZYME OR PROCESS AFFECTED
MSH2, 3, 6, MLH1, PMS2	colon cancer	mismatch repair
Xeroderma pigmentosum (XP) groups A–G	skin cancer, cellular UV sensitivity, neurological abnormalities	nucleotide excision-repair
XP variant	cellular UV sensitivity	translesion synthesis by DNA polymerase δ
Ataxia–telangiectasia (AT)	leukemia, lymphoma, cellular γ-ray sensitivity, genome instability	ATM protein, a protein kinase activated by double-strand breaks
BRCA-2	breast and ovarian cancer	repair by homologous recombination
Werner syndrome	premature aging, cancer at several sites, genome instability	accessory 3′-exonuclease and DNA helicase
Bloom syndrome	cancer at several sites, stunted growth, genome instability	accessory DNA helicase for replication
Fanconi anemia groups A–G	congenital abnormalities, leukemia, genome instability	DNA interstrand cross-link repair
46 BR patient	hypersensitivity to DNA-damaging agents, genome instability	DNA ligase I

There are many more, we only talk about a few of them in this course

Most are Indirect Repair

- **Main Mechanisms**
  - *But not intended to be comprehensive*
- **Nucleotide Excision**
  - -removes more than a few (up to 30) base around a damaged site
  - This is how pyrimidine dimers formed by UV damage are repaired
- **Base Excision**
  - repairs a single (or just a few) damaged bases by removing it (damage by methylation, oxidation, *etc*.)
- **Mismatch Repair** (post replication repair)
  - repairs mismatched bases
  - that formed due to tautomerism for *e.g*.)

Excision Repair, Mechanism

Mechanism is essentially the same for all excision repair, just the size of the excised region varies

Involves

- recognition of damage
- removal of damaged base or region around damaged base

specific repair proteins

- replacement of excised region

shared repair proteins

Excision Repair

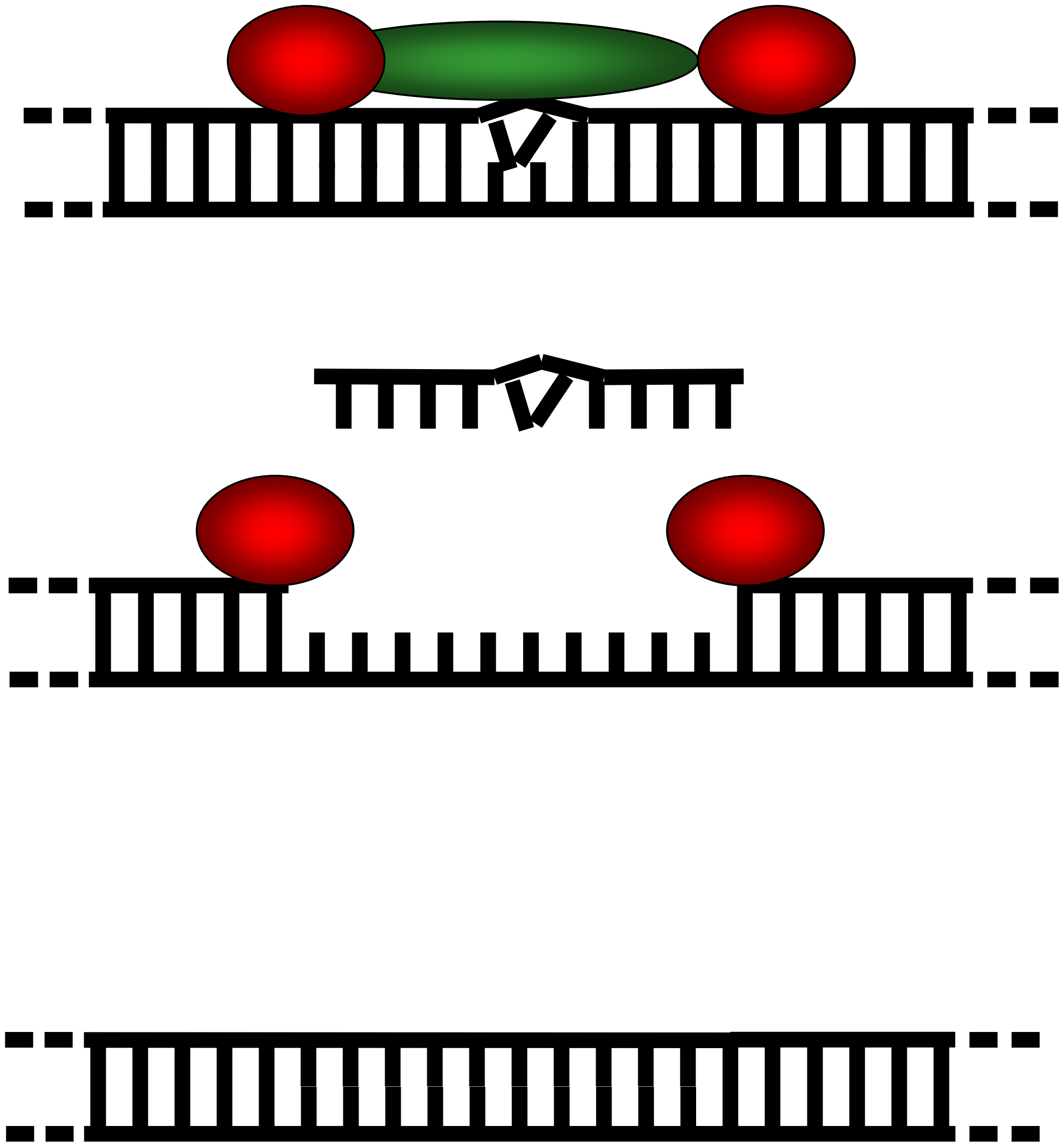
1) Recognition of damage

2) Recruit endonucleases

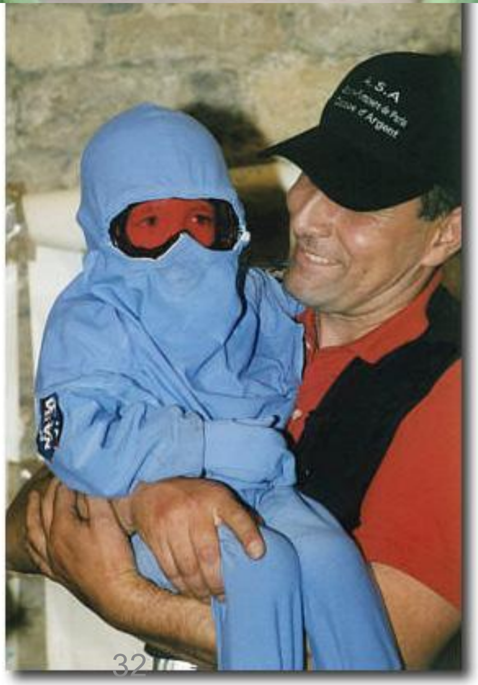
3) Region excised

4) DNA Polymerase fills in gap

5) Ligase seal nick



Xeroderma pigmentosum (XP)



<u>Autosomal Recessive</u>, biallelic pathogenic mutations in 9 different **NER** genes can lead to XP

- So 9 complementation groups
- *Locus heterogeneity because there are many genes involved in the NER pathway*

- *XPA*, *ERCC3 (XPB)*, *XPC*, *ERCC2 (XPD)*, *DDB2 (XPE)*, *ERCC4 (XPF)*, *ERCC5 (XPG)*, *ERCC1,* and *POLH (XP-V)*.
- *Recognize that* ***XP*** *is the common gene name for NER pathway components; other names do not need to be memorized)*

-incidence

- Europe-USA, 1:250,000
- Japan, 1:40,000

= extreme sun sensitivity

Xeroderma pigmentosum (XP)

-clinical sun sensitivity,

= sunburn, blistering, freckled with hyperpigmented skin lesions

-ocular involvement

= conjunctivitis, ocular tumours

> 1000-fold increase in skin cancer including melanomas

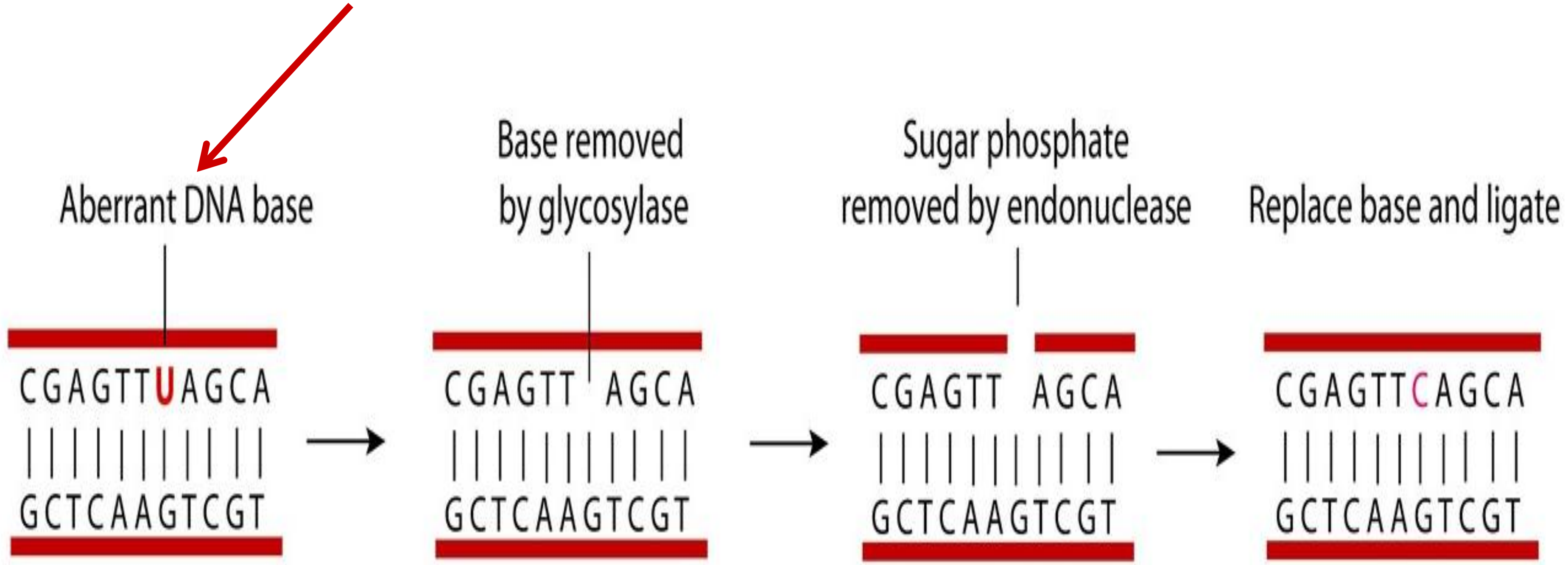
-20% have progressive neurologic degeneration

-DNA damage is cumulative and of course, irreversible

Base Excision Repair

Damaged nucleotides can be removed by DNA glycosylases which recognize specific damaged bases in DNA.

Uracil glycosylase as a specific example for removal of U from DNA



There are many different types of DNA glycosylases which are specific for different aberrant bases

Repair mismatched bases

Mismatch repair

= post-replicative repair mechanism

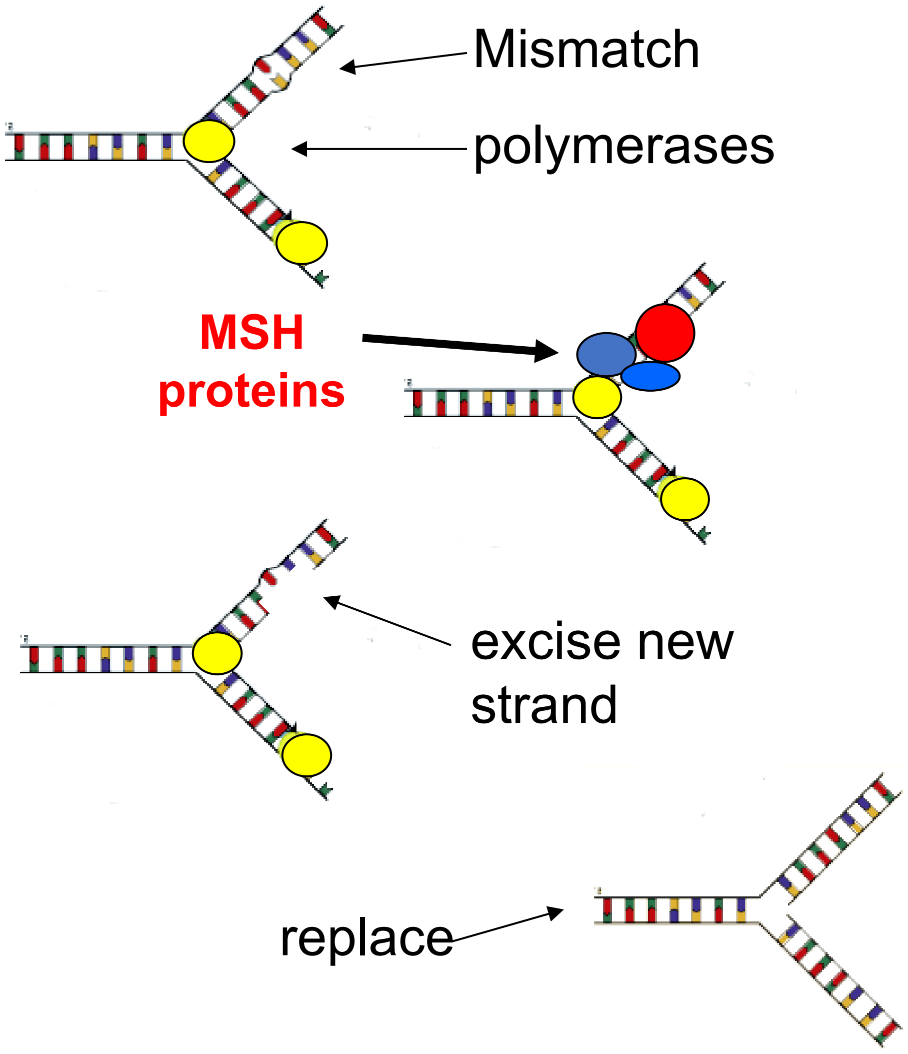
-a form of excision repair (same basic mechanism, just different proteins involved)

-mismatched bases are recognized and excised

-also very important in relation to removing small repeats that tend to expand (*e.g.* triplet expansion disorders)

Mismatch Repair (a type of excision repair) shows strand discrimination

1) Mismatch missed by proofreading is recognized by MSH proteins
2) Repair may occur during S-phase (if missed by proof-reading) or in G2 when genome is scanned for errors
3) Excision of bases around mismatch
4) Repair by re-synthesis

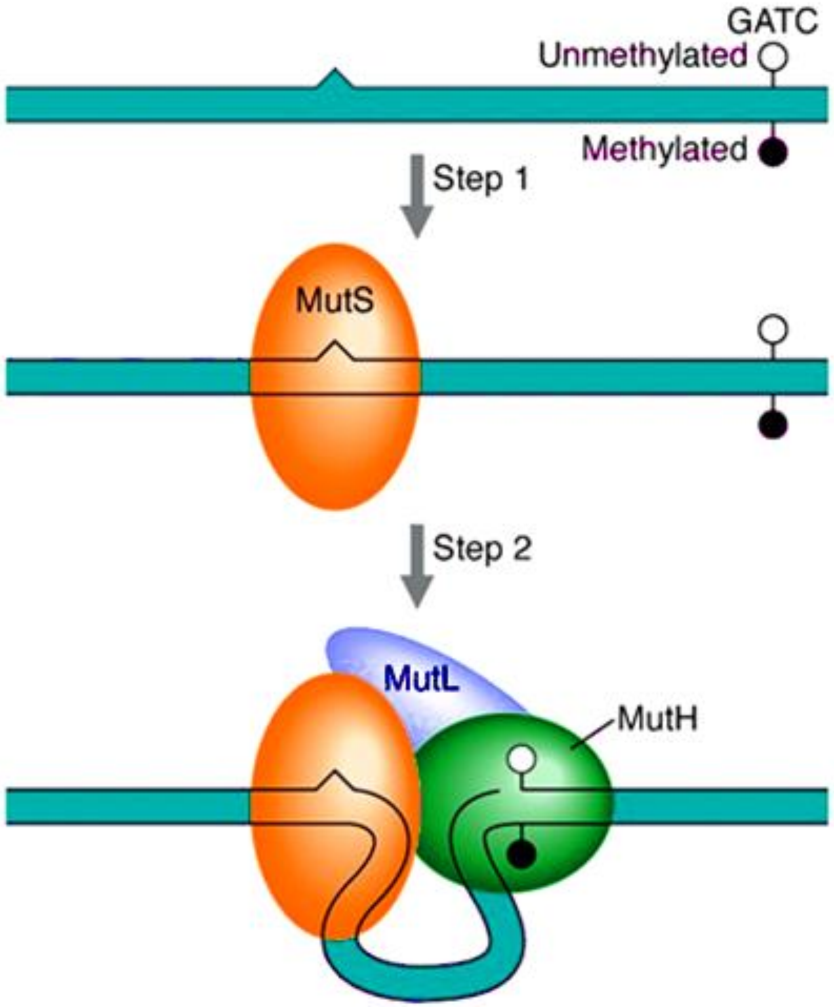


Many genes function in this pathway such as *MSH2/6, MLH1-3, PMS2 + several more*

(it is enough to know that the **MSH** genes are the founding members = **m**is**s**-matc**h**)

Strand Discrimination
How does the cell know which is the "right" strand

In prokaryotes it has been shown that hemimethylated strands are recognized. The methylated strand is the 'original" and the unmethylated strand is the new one



In eukaryotes it may be more complex (of course) and interaction with replication machinery occurs but it also involves discernment of methylated / unmethylated DNA strands

Hereditary Nonpolyposis Colon Cancer (autosomal dominant)

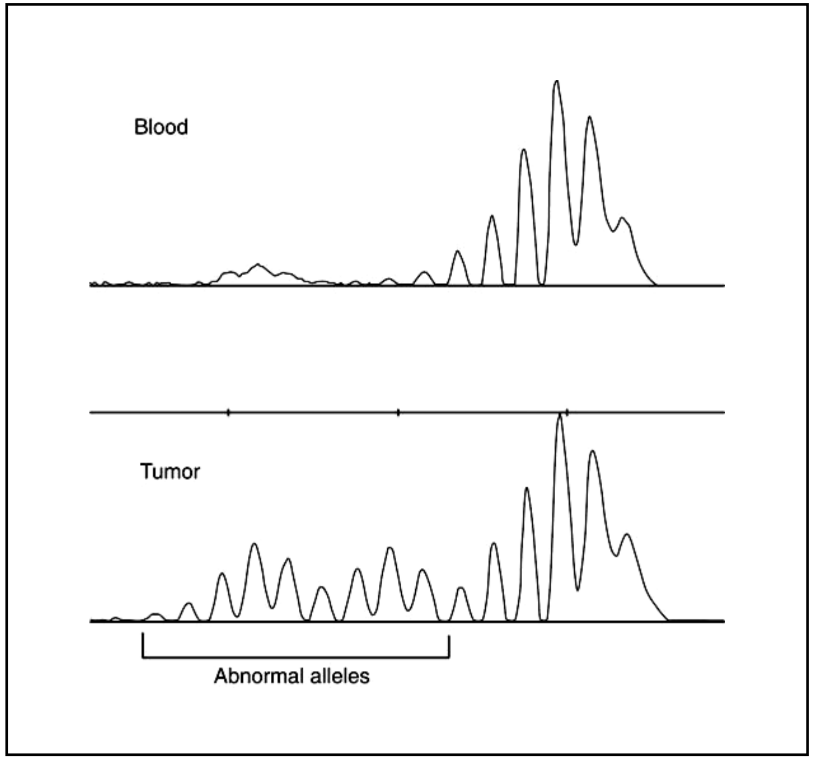
-a result of mutations in genes encoding mismatch repair proteins ***MSH***_2*_, *MLH1**, *PMS1*, *PMS2*, or *MSH6*

(*most common mutations)

-results in **microsatellite instability**

Microsatellite instability frequently seen with these tumours
= simple repetitive DNA tandem repeat sequences show size variability due to inaccurate replication

SSR



See also Korf, Fig. 8.15

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Double stranded breaks: a difficult type of mutation to repair.

DSBs are particularly dangerous to dividing cells; high probability of loss of genetic material or chromosomal rearrangement

Two mechanisms available to deal with this:

- non-homologous end joining
  - -the more common method used
  - -does not use homologous chromosome to repair the break
- recombinational repair
  - -does use homologous chromosome
  - -less error prone than NHEJ

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BRCA1 and BRCA 2 (autosomal dominant)

= Autosomal dominant breast cancer predisposition syndrome

-Found in the cells of breast and other tissue

-Involved in DNA repair or apoptosis when DNA can't be repaired.

-Women with *BRCA1* or *BRCA2* mutations have up to an 85% risk of developing breast cancer by age 70

-And the risk of developing ovarian cancer is about 55% for *BRCA1* and 25% for *BRCA2* mutations

-This cancer risk increases as the person ages (of course)

-Two different genes, so locus heterogeneity

-Many hundreds of pathogenic variants have been identified in the *BRCA1* gene, most associated with an increased risk of cancer (*allelic heterogeneity)*

Despite all of this some errors get through

Somatic errors

-can result in cancer

-does result in aging

bad for you

Germline errors

-can result in genetic diseases visited on your descendants

bad for your kids

but on the good side, DNA damage provides diversity for evolution

On average, it is thought that in gametes, one mutation occurs per ~10^8 nucleotides replicated, or ~60 mutations occur in every cell generation. (human genome is 6 x 10^9 nucleotides)

DNA repair genes are also vulnerable to mutation

Consequences may be very serious

1) **Increased error rate**
    - Particularly true for proofreading and mismatch repair (MMR), produces “mutator” phenotype
    - Damaged bases are not repaired efficiently
2) **Genomic Instability**
    - mutations in gene involved in resting DNA repair and chromosome break repair will de-stabilize the genome

      e.g. Bloom syndrome, XP, many others

Ataxia telangiectasia

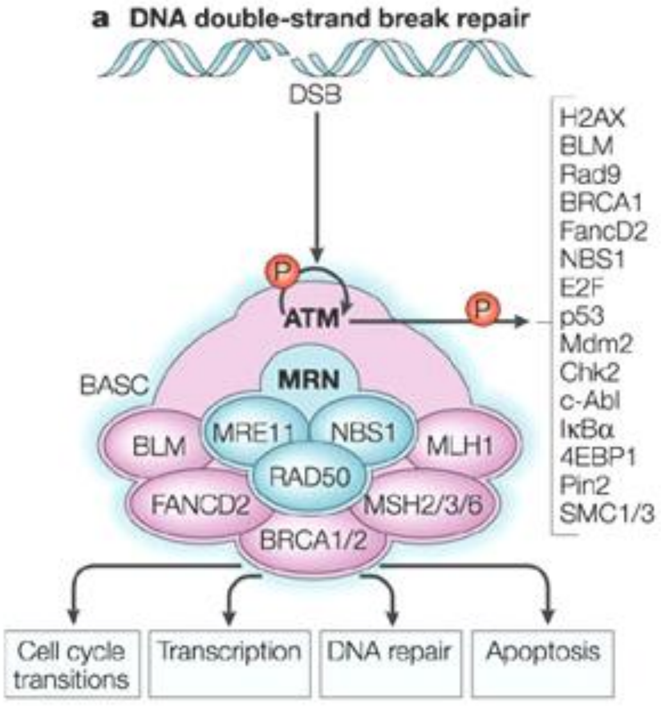
defect is in *ATM* gene (11q22-23)

= A serine-threonine protein kinase with a number of functions including:

-detecting DNA damage (*i.e*. it is a sensor) and activating cell cycle arrest and DNA repair proteins (*e.g*. p53)

-autosomal recessive inheritance

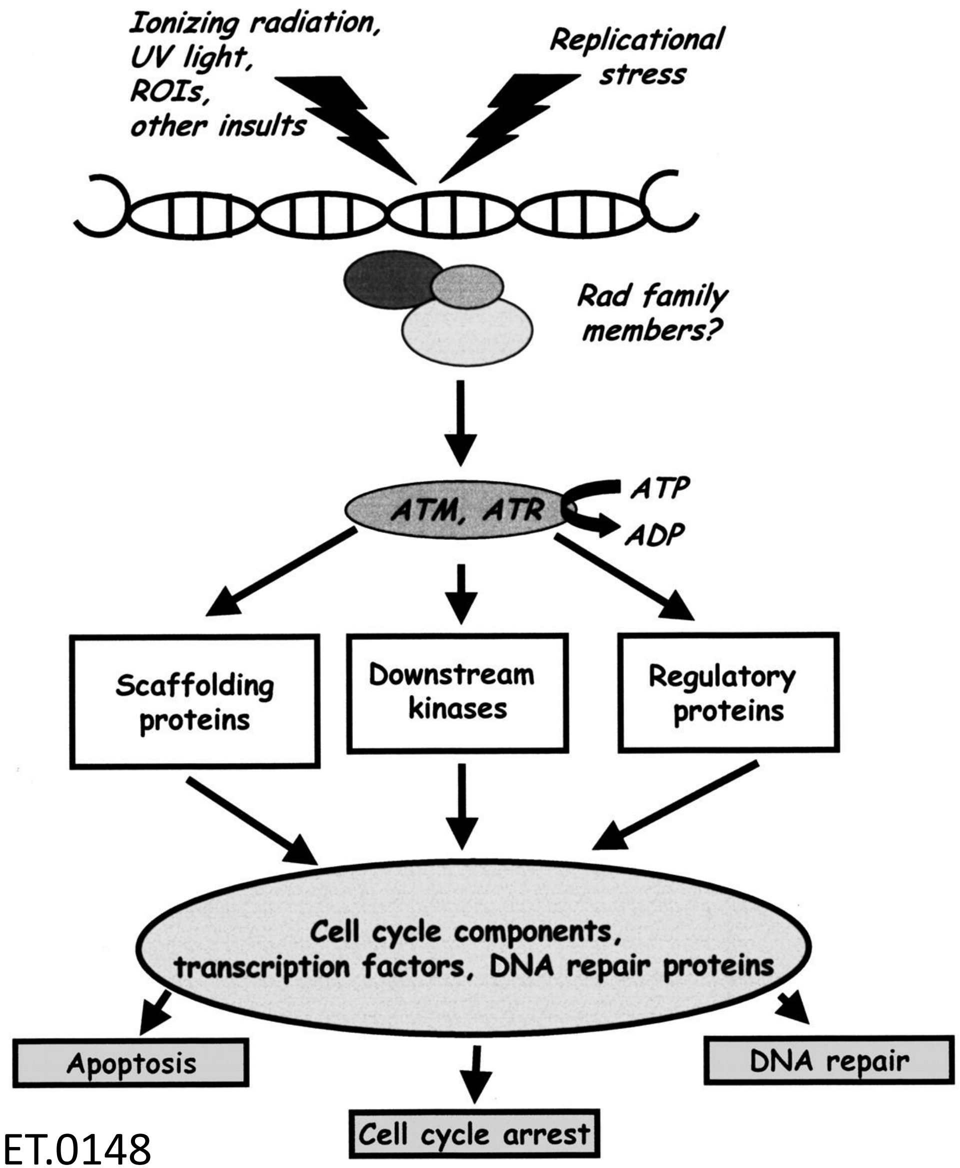
-rare disease (incidence estimated to be 1:50,000)

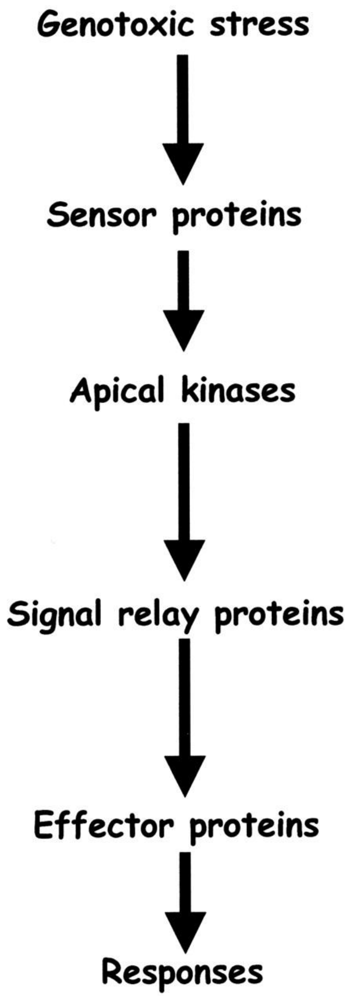


-increased cancer risk (should avoid x-rays)
affects cerebellum (= ataxia) and immune system
--ocular telangiectasia common

Fig. 18-23. Ocular telangiectasia in a child with ataxia telangiectasia.

ATM/ATR-mediated signaling pathway

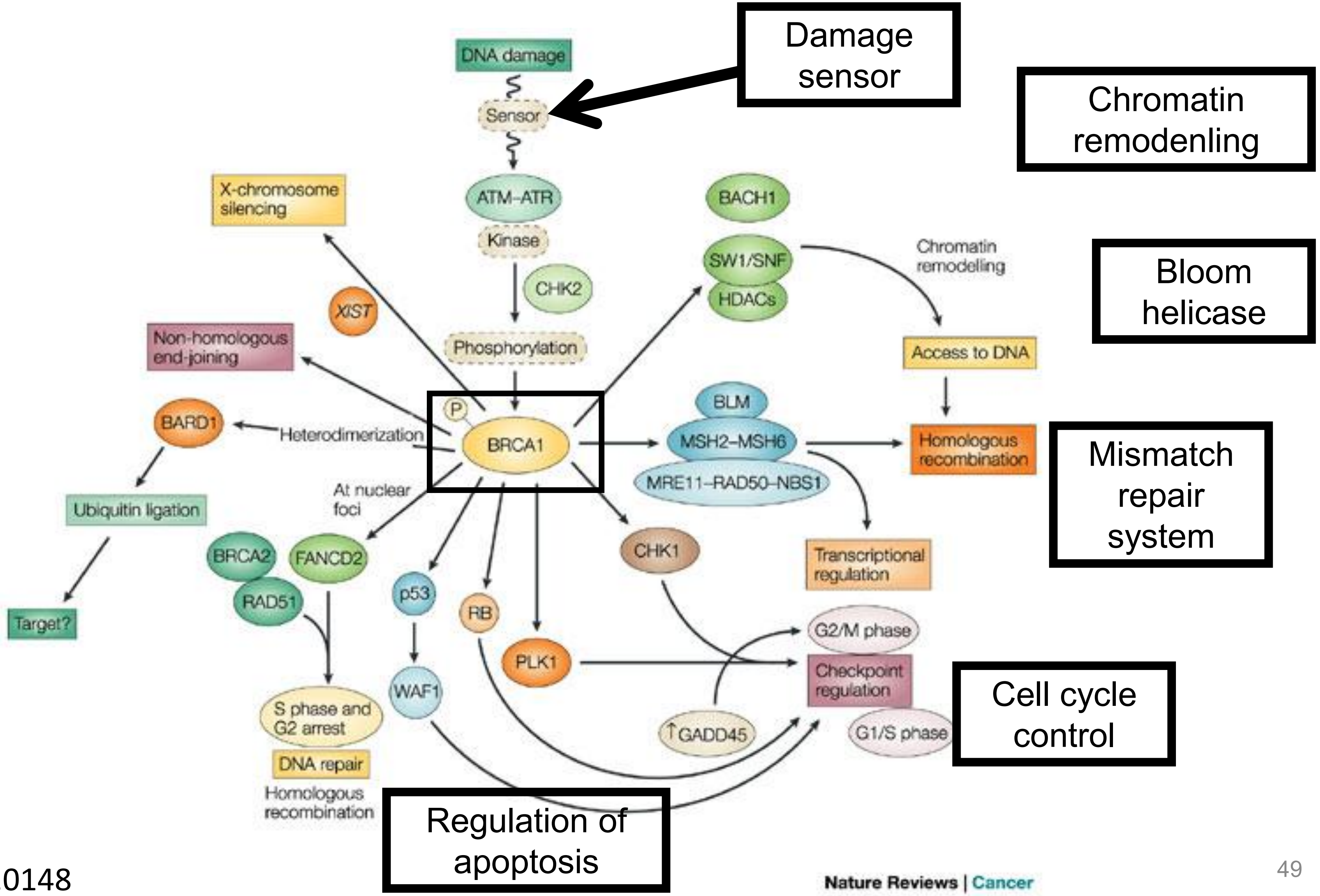




Most of the DNA repair pathways interact with, and influence each other. DNA repair, cell cycle control, and cell survival / apoptosis are all interconnected

This figure can be redrawn many different ways to put different DNA repair components in the center (as central controllers)

Main point – they all interact together to carry out their cellular functions



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Key Points

- Patients with DNA repair defects may have developmental delay, neurocognitive issues, and extreme sensitivity to DNA damaging agents
- Patients with DNA repair deficiencies should avoid DNA damaging agents
  - Exposure to UV light
  - X-rays, CT scan, etc
  - Radiation therapy
  - Some chemotherapy agents which works through DNA damage with the objective to initiate apoptosis (i.e. cisplatin)
- Patients with DNA repair defects usually have a higher risk for developing cancer
- DNA repair is linked with carcinogenesis and apoptosis

Key points expanded for use as a study aid: DNA Repair and Associated Disorders

- In this session we discussed how mutations arise, and some of the repair mechanisms that reduce mutation frequency. We also describe several disorders associated with defects of DNA repair and the general indications associated with those disorders.

- Predict the short- and long-term consequences of unrepaired DNA damage
- Give examples of the common types of DNA damage that lead to mutations
- Discuss the formation of UV light photoproducts and in particular thymine dimers
- Describe and distinguish DNA repair mechanisms including:
  - Proof reading
  - Base and nucleotide excision repair
  - Mismatch repair
  - Non-homologous end joining
- Explain the importance of the complementary strand as a template for repair synthesis
- Explain the possible consequences of mutations in the DNA repair genes:
  - Genomic instability
  - Hereditary DNA repair disorders (Xeroderma pigmentosum, Bloom syndrome, Ataxia telangiectasia). Mention the genetic defect in each of the disorders listed
- Identify, compare, and contrast the genes implicated in hereditary nonpolyposis colon cancer (HNPCC), Familial adenomatous colon cancer (FAP), and familial breast cancer (BrCa)